

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

If the Coefficient of x^{30} in the expansion of $\left(1 + \frac{1}{x}\right)^6 (1 + x^2)^7 (1 - x^3)^8$; $x \neq 0$ is α , then $|\alpha|$ equals

Q2 - 2024 (01 Feb Shift 2)

Let m and n be the coefficients of seventh and thirteenth terms respectively in the expansion of

$\left(\frac{1}{3}x^{\frac{1}{3}} + \frac{1}{2x^{\frac{2}{3}}}\right)^{18}$. Then $\left(\frac{n}{m}\right)^{\frac{1}{3}}$ is :

- (1) $\frac{4}{9}$
- (2) $\frac{1}{9}$
- (3) $\frac{1}{4}$
- (4) $\frac{9}{4}$

Q3 - 2024 (27 Jan Shift 1)

${}^{n-1}C_r = (k^2 - 8) {}^nC_{r+1}$ if and only if :

- (1) $2\sqrt{2} < k \leq 3$
- (2) $2\sqrt{3} < k \leq 3\sqrt{2}$
- (3) $2\sqrt{3} < k < 3\sqrt{3}$
- (4) $2\sqrt{2} < k < 2\sqrt{3}$

Q4 - 2024 (27 Jan Shift 1)

If A denotes the sum of all the coefficients in the expansion of $(1 - 3x + 10x^2)^n$ and B denotes the sum of all the coefficients in the expansion of $(1 + x^2)^n$, then :

- (1) $A = B^3$
- (2) $3A = B$
- (3) $B = A^3$
- (4) $A = 3B$

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Q5 - 2024 (27 Jan Shift 2)

The coefficient of x^{2012} in the expansion of $(1-x)^{2008}(1+x+x^2)^{2007}$ is equal to

Q6 - 2024 (29 Jan Shift 1)

If $\frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \dots + \frac{{}^{11}C_9}{10} = \frac{n}{m}$ with $\gcd(n, m) = 1$, then $n + m$ is equal to

Q7 - 2024 (29 Jan Shift 2)

Remainder when $64^{32^{32}}$ is divided by 9 is equal to _____.

Q8 - 2024 (30 Jan Shift 1)

Number of integral terms in the expansion of $\left\{7^{\left(\frac{1}{2}\right)} + 11^{\left(\frac{1}{6}\right)}\right\}^{824}$ is equal to _____

Q9 - 2024 (30 Jan Shift 2)

Suppose $2 - p, p, 2 - \alpha, \alpha$ are the coefficient of four consecutive terms in the expansion of $(1+x)^n$. Then the value of $p^2 - \alpha^2 + 6\alpha + 2p$ equals

[We changed options. In official NTA paper no option was correct.]

(1) 4

(2) 10

(3) 8

(4) Data Inconsistent

Q10 - 2024 (30 Jan Shift 2)

Let $\alpha = \sum_{k=0}^n \left(\frac{{}^nC_k}{k+1} \right)$ and $\beta = \sum_{k=0}^{n-1} \left(\frac{{}^nC_k {}^nC_{k+1}}{k+2} \right)$

If $5\alpha = 6\beta$, then n equals

Q11 - 2024 (31 Jan Shift 1)

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In the expansion of $(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$, $x \neq 0$, the sum of the coefficient of x^3 and x^{-13} is equal to _____.

Q12 - 2024 (31 Jan Shift 2)

If for some m, n ; ${}^6C_m + 2({}^6C_{m+1}) + {}^6C_{m+2} > {}^8C_3$ and ${}^{n-1}P_3 : {}^nP_4 = 1 : 8$, then ${}^nP_{m+1} + {}^{n+1}C_m$ is equal to

(1) 380

(2) 376

(3) 384

(4) 372

Q13 - 2024 (31 Jan Shift 2)

Let the coefficient of x^r in the expansion of

$$(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \dots + (x+2)^{n-1}$$

be α_r . If $\sum_{r=0}^n \alpha_r = \beta^n - \gamma^n$, $\beta, \gamma \in N$, then the value of $\beta^2 + \gamma^2$ equals _____.

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Answer Key

Q1 (678)	Q2 (4)	Q3 (1)	Q4 (1)
Q5 (0)	Q6 (2041)	Q7 (1)	Q8 (138)
Q9 (4)	Q10 (10)	Q11 (118)	Q12 (4)
Q13 (25)			

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Solutions

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Q1

coeff of x^{30} in $\frac{(x+1)^6(1+x^2)^7(1-x^3)^8}{x^6}$ coeff. of x^{36} in $(1+x)^6(1+x^2)^7(1-x^3)^8$

General term

$${}^6C_{r_1} {}^7C_{r_2} {}^8C_{r_3} (-1)^{r_3} X^{r_1+2r_2+3r_3}$$

$$r_1 + 2r_2 + 3r_3 = 36$$

Case-I :

r_1	r_2	r_3
0	6	8
2	5	8
4	4	8
6	3	8

$$r_1 + 2r_2 = 12 \text{ (Taking } r_3 = 8)$$

r_1	r_2	r_3
1	7	7
3	6	7
5	5	7

Case II.

$$r_1 + 2r_2 = 15 \text{ (Taking } r_3 = 7)$$

r_1	r_2	r_3
4	7	6
6	6	6

Case-III :

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Solutions

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$$r_1 + 2r_2 = 18 \text{ (Taking } r_3 = 6 \text{)}$$

$$\text{Coeff.} = 7 + (15 \times 21) + (15 \times 35) + (35)$$

$$- (6 \times 8) - (20 \times 7 \times 8) - (6 \times 21 \times 8) + (15 \times 28)$$

$$+ (7 \times 28) = -678 = \alpha$$

$$|\alpha| = 678$$

Q2

$$\left(\frac{x^{\frac{1}{3}}}{3} + \frac{x^{\frac{-2}{3}}}{18} \right)^{18}$$

$$t_7 = {}^{18}C_6 \left(\frac{x^{\frac{1}{3}}}{3} \right)^{12} \left(\frac{x^{\frac{-2}{3}}}{2} \right)^6 = {}^{18}C_6 \frac{1}{(3)^{12}} \cdot \frac{1}{2^6}$$

$$t_{13} = {}^{18}C_{12} \left(\frac{x^{\frac{1}{3}}}{3} \right)^6 \left(\frac{x^{\frac{-2}{3}}}{2} \right)^{12} = {}^{18}C_{12} \frac{1}{(3)^6} \cdot \frac{1}{2^{12}} \cdot x^{-6}$$

$$m = {}^{18}C_6 \cdot 3^{-12} \cdot 2^{-6} : n = {}^{18}C_{12} \cdot 2^{-12} \cdot 3^{-6}$$

$$\left(\frac{n}{m} \right)^{\frac{1}{3}} = \left(\frac{2^{-12} \cdot 3^{-6}}{3^{-12} \cdot 2^{-6}} \right)^{\frac{1}{3}} = \left(\frac{3}{2} \right)^2 = \frac{9}{4}$$

Q3

$${}^{n-1}C_r = (k^2 - 8) {}^nC_{r+1}$$

$$\underbrace{r+1 \geq 0, r \geq 0}_{r \geq 0}$$

$$\frac{{}^nC_r}{{}^nC_{r+1}} = k^2 - 8$$

$$\frac{r+1}{n} = k^2 - 8$$

$$\Rightarrow k^2 - 8 > 0$$

$$(k - 2\sqrt{2})(k + 2\sqrt{2}) > 0$$

$$k \in (-\infty, -2\sqrt{2}) \cup (2\sqrt{2}, \infty) \dots (I)$$

$$\therefore n \geq r+1, \frac{r+1}{n} \leq 1$$

$$\Rightarrow k^2 - 8 \leq 1$$

$$k^2 - 9 \leq 0$$

$$-3 \leq k \leq 3 \dots (II)$$

From equation (I) and (II) we get

$$k \in [-3, -2\sqrt{2}) \cup (2\sqrt{2}, 3]$$

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Solutions

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Q4

Sum of coefficients in the expansion of

$$(1 - 3x + 10x^2)^n = A$$

$$\text{then } A = (1 - 3 + 10)^n = 8^n \text{ (put } x = 1)$$

and sum of coefficients in the expansion of

$$(1 + x^2)^n = B$$

$$\text{then } B = (1 + 1)^n = 2^n$$

$$A = B^3$$

Q5

$$(1 - x)(1 - x)^{2007}(1 + x + x^2)^{2007}$$

$$(1 - x)(1 - x^3)^{2007}$$

$$(1 - x) \left({}^{2007}C_0 - {}^{2007}C_1(x^3) + \dots \right)$$

General term

$$(1 - x) \left((-1)^r {}^{2007}C_r x^{3r} \right)$$

$$(-1)^r {}^{2007}C_r x^{3r} - (-1)^r {}^{2007}C_r x^{3r+1}$$

$$3r = 2012$$

$$r \neq \frac{2012}{3}$$

$$3r + 1 = 2012$$

$$3r = 2011$$

$$r \neq \frac{2011}{3}$$

Hence there is no term containing x^{2012} .So coefficient of $x^{2012} = 0$

Q6

$$\sum_{r=1}^9 \frac{{}^{11}C_r}{r+1}$$

$$= \frac{1}{12} \sum_{r=1}^9 {}^{12}C_{r+1}$$

$$= \frac{1}{12} [2^{12} - 26] = \frac{2035}{6}$$

$$\therefore m + n = 2041$$

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Solutions

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Q7

$$\begin{aligned}\text{Let } 32^{32} &= t \\ 64^{32^{32}} &= 64^t = 8^{2t} = (9-1)^{2t} \\ &= 9k + 1 \\ \text{Hence remainder} &= 1\end{aligned}$$

Q8

General term in expansion of $((7)^{1/2} + (11)^{1/6})^{824}$ is

$$t_{r+1} = {}^{824}C_r (7)^{\frac{824-r}{2}} (11)^{r/6}$$

For integral term, r must be multiple of 6.

Hence $r = 0, 6, 12, \dots, 822$

Q9

No solution. Refer to answer key.

Q10

$$\alpha = \sum_{k=0}^n \frac{{}^nC_k \cdot {}^nC_k}{k+1} \cdot \frac{n+1}{n+1}$$

$$= \frac{1}{n+1} \sum_{k=0}^n {}^{n+1}C_{k+1} \cdot {}^nC_{n-k}$$

$$\alpha = \frac{1}{n+1} \cdot {}^{2n+1}C_{n+1}$$

$$\beta = \sum_{k=0}^{n-1} {}^nC_k \cdot \frac{{}^nC_{k+1}}{k+2} \cdot \frac{n+1}{n+1}$$

$$= \frac{1}{n+1} \sum_{k=0}^{n-1} {}^nC_{n-k} \cdot {}^{n+1}C_{k+2}$$

$$= \frac{1}{n+1} \cdot {}^{2n+1}C_{n+2}$$

$$\frac{\beta}{\alpha} = \frac{2n+1}{2n+1} C_{n+2}$$

$$\frac{\beta}{\alpha} = \frac{2n+1 - (n+2) + 1}{n+2} = \frac{5}{6}$$

$$n = 10$$

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Solutions

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Q11

$$\begin{aligned}
 & (1+x)(1-x^2) \left(1 + \frac{3}{x} + \frac{3}{x^2} + \frac{1}{x^3}\right)^5 \\
 &= (1+x)(1-x^2) \left(\left(1 + \frac{1}{x}\right)^3\right)^5 \\
 &= \frac{(1+x)^2(1-x)(1+x)^{15}}{x^{15}} \\
 &= \frac{(1+x)^{17} - x(1+x)^{17}}{x^{15}} \\
 &= \text{coeff } (x^3) \text{ in the expansion} \approx \text{coeff } (x^{18}) \text{ in } \\
 & (1+x)^{17} - x(1+x)^{17} \\
 &= 0 - 1 \\
 &= -1 \\
 & \text{coeff } (x^{-13}) \text{ in the expansion} \approx \text{coeff } (x^2) \text{ in } \\
 & (1+x)^{17} - x(1+x)^{17} \\
 &= \binom{17}{2} - \binom{17}{1} \\
 &= 17 \times 8 - 17 \\
 &= 17 \times 7 \\
 &= 119 \\
 & \text{Hence Answer} = 119 - 1 = 118
 \end{aligned}$$

Q12

$$\begin{aligned}
 & {}^6C_m + 2({}^6C_{m+1}) + {}^6C_{m+2} > {}^8C_3 \\
 & {}^7C_{m+1} + {}^7C_{m+2} > {}^8C_3 \\
 & {}^8C_{m+2} > {}^8C_3 \\
 & \therefore m = 2 \\
 & \text{And } {}^{n-1}P_3 : {}^nP_4 = 1 : 8 \\
 & \frac{(n-1)(n-2)(n-3)}{n(n-1)(n-2)(n-3)} = \frac{1}{8} \\
 & \therefore n = 8 \\
 & \therefore {}^nP_{m+1} + {}^{n+1}C_m = {}^8P_3 + {}^9C_2 \\
 &= 8 \times 7 \times 6 + \frac{9 \times 8}{2} \\
 &= 372
 \end{aligned}$$

Q13

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Solutions

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$$\begin{aligned}
 & (x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3} \\
 & (x+2)^2 + \dots + (x+2)^{n-1} \\
 \sum \alpha_r &= 4^{n-1} + 4^{n-2} \times 3 + 4^{n-3} \times 3^2 + \dots + 3^{n-1} \\
 &= 4^{n-1} \left[1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \dots + \left(\frac{3}{4}\right)^{n-1} \right] \\
 &= 4^{n-1} \times \frac{1 - \left(\frac{3}{4}\right)^n}{1 - \frac{3}{4}} \\
 &= 4^n - 3^n = \beta^n - \gamma^n \\
 \beta &= 4, \gamma = 3 \\
 \beta^2 + \gamma^2 &= 16 + 9 = 25
 \end{aligned}$$

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